O-C-105a
SEPTEMBER 18, 1958
SUPERSEDING
Int. Fed. Spec. O–C–00105 (GSA–FSS)
June 28, 1957
(See 6.4)

FEDERAL SPECIFICATION

CALCIUM CHLORIDE, DIHYDRATE AND CALCIUM CHLORIDE, ANHYDROUS; TECHNICAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification is intended to establish the requirements for the technical grades of hydrated and anhydrous calcium chloride (see 6.1).

1.2 Classification

1.2.1 *Types, grades, and classes.*—The calcium chloride covered by this specification shall be of the following types, grades, and classes, as specified:

Type I.—Dihydrate, 74 percent, flake.
Type II.—Anhydrous.
Grade A.—94 percent, flake or pellet.
Grade B.—Desiccant.
Class 1.—Granular.
Class 2.—Lump.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

U–M–186—Medicinal Products and Clinical Laboratory Reagents; General Specification for Containers (Packaging and Packing).
UU–S–48—Sacks, Paper, Shipping.
PPP–B–35 — Bags: Textile, Shipping, Burlap, Cotton and Waterproof Laminated.
PPP–D–723—Drums; Fiber.
PPP–D–729 — Drums; Metal, 55-Gallon (For Shipment of Noncorrosive Materials).
PPP–S–50—Sacks, Shipping, Paper, Reinforced.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 536—Soap and Soap Products (Including Synthetic Detergents); Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

FSC 6810

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:12Z
Check the source to verify that this is the current version before use.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. — The following documents form a part of this specification. Unless otherwise specified, the issue in effect on date of invitation for bids shall apply.

ASTM Standards—Part 3

ASTM Designation: D 345 — Standard Methods of Sampling and Testing Calcium Chloride.

(Copies may be obtained from the American Society for Testing Materials, 1916 Race Street, Philadelphia, Pennsylvania.)

Uniform Freight Classification:

Ratings, Rules and Regulations

(Copies may be obtained from the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also widely distributed among technical groups and using Federal agencies.

2.2.1 The requirements for type I and type II, grade A calcium chloride are essentially similar to the requirements of ASTM Designation: D 98 – 56 T, Tentative Specifications for Calcium Chloride.

3. REQUIREMENTS

3.1 Material.—Calcium chloride shall not be caked or sticky in the container when received and shall conform to the chemical requirements shown in table I.

3.2 Particle size.—The material shall conform to the requirements shown in table II when tested as specified in 4.3.1.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.

4.1.1 *Lot.*—For the purposes of sampling, a lot shall consist of all the calcium chloride of the same type, grade, and class manufactured as one batch and offered for delivery at one time.

4.1.1.1 *Batch.*—A batch shall be that quantity of the chemical manufactured at the same time. When practicable, all calcium chloride of the same type, grade, and class supplied on one order shall be from the same manufacturer's batch.

4.1.2 *Sampling for inspection of filled containers.*—A random sample of filled containers shall be selected from each lot in accordance with Military Standard MIL–STD–105 at Inspection Level I, and Acceptable Quality Level (AQL) = 2.5 percent defective to verify compliance with all stipulations of this specification regarding fill, closure, marking, and other requirements not involving tests.

4.1.3 *Sampling for tests.*—From each inspection lot, two containers shall be selected at random. From each of the two containers enough material for tests shall be taken and placed in separate, clean, dry glass containers, sealed, marked, and forwarded to the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:12Z
Check the source to verify that this is the current version before use.

TABLE I.—*Chemical requirements*

	Type I	Type II		Test paragraph
		Grade A	Grade B	
Calcium chloride, minimum, percent	[1] 74.0	94.0	90.0	4.3
Total alkali chloride (as NaCl), maximum, percent	2.0	5.0	4.2	4.3
Total magnesium (as $MgCl_2$), maximum, percent	0.5	0.5	0.5	4.3

[1] Where factory inspection is carried out, the calcium chloride content shall be not less than 77.0 percent.

TABLE II.—*Sieve analysis*

Sieve size or number	Type I	Type II		
		Grade A	Grade B	
			Class 1	Class 2
Passing 1-inch sieve				100 percent
Passing ⅜-inch sieve	100 percent	100 percent	100 percent	
Passing U. S. No. 4	80 percent, minimum	80 percent, minimum	90 percent, minimum	5 percent, maximum
Passing U. S. No. 8			50 percent, maximum	
Passing U. S. No. 16			5 percent, maximum	
Passing U. S. No. 20	10 percent, maximum	10 percent, maximum		

testing laboratory designated by the activity concerned.

4.1.3.1 Each of the containers shall be sampled by scraping aside the top layer to a depth of approximately 1-inch and taking samples by means of a sampling thief or other method which will insure a sample that is representative of the material in the container to a depth of at least six inches. Precautions should be taken during the sampling operation to avoid exposing the sample unduly to atmospheric moisture.

4.2 Inspection.

4.2.1 *Inspection of filled container.*—Each sample filled container selected in accordance with 4.1.2 shall be examined by the Government inspector for defects of the container

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:12Z
Check the source to verify that this is the current version before use.

and closure, for evidence of leakage, and for unsatisfactory markings. Each sample filled container shall also be weighed to determine the amount of the contents. Any container in the sample having one or more defects, or under fill shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of Military Standard MIL–STD–105, the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for examination provided that the contractor has removed or repaired all nonconforming containers.

4.2.2 *Inspection tests.*—The sample specimens selected in accordance with 4.1.3 shall be subjected separately to the tests specified in 4.3. If either specimen fails in one or more of these tests, the lot shall be rejected. Rejected lots may be resubmitted for acceptance provided the contractor has removed or reworked all nonconforming products.

4.3 **Tests.**—Tests shall be made in accordance with the applicable method of ASTM Designation: D 345 and as specified hereinafter.

4.3.1 *Particle size.* — Sieve tests shall be made in accordance with applicable sections of method 2101 of Federal Test Method Standard No. 536. The sieves being used shall be nested with pan and top, and sieving shall be completed as rapidly as possible.

5. PREPARATION FOR DELIVERY

For civil agency procurement, definitions and applications of the levels of packaging and packing shall be in accordance with Federal Standard No. 102.

5.1 Packaging.

5.1.1 *Level A.*—Calcium chloride shall be packaged in 1-pound or 5-pound glass bottles in accordance with Federal Specification U–M–186, except when bulk packing is specified (see 5.2.1).

5.1.2 *Level C.*—The calcium chloride shall be packaged in units of 1-pound in glass containers equipped with suitable closures or in units of 5 pounds in suitable containers in accordance with the best commercial practice which will prevent moisture absorption, except when bulk packing is specified (see 5.2.2 and 5.2.3).

5.2 Packing.

5.2.1 *Level A.*—Calcium chloride shall be packed in laminated textile bags, multiwall sacks, or steel containers as follows:

5.2.1.1 *Multiwall sacks.*—Multiwall sacks shall conform to the requirements of Federal Specification PPP–S–50.

5.2.1.2 *Steel containers.*—Steel containers shall conform to the requirements of Federal Specification PPP–D–729. The gross weight shall not exceed 700 pounds.

5.2.1.3 *One-pound and five-pound containers.*—One-pound and five-pound containers shall be packed in accordance with the overseas requirements of Federal Specification U–M–186.

5.2.2 *Level B.*—Calcium chloride shall be packed in fiber drums, laminated textile bags, multiwall sacks, or steel containers as follows:

5.2.2.1 *Fiber drums.*—Fiber drums shall conform to the requirements of Federal Specification PPP–D–723, type I or II, grade A. The drums shall be furnished with a moisture-vapor barrier, lining, interior coating, or polyethylene bag giving the required protection for hygroscopic materials. The gross weight shall not exceed approximately 100 pounds.

5.2.2.2 *Laminated textile bags.*—50- or 100-pound quantities of calcium chloride shall be packed in laminated textile bags conforming to the requirements of Federal Specification PPP–B–35, table XI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:12Z
Check the source to verify that this is the current version before use.

5.2.2.3 *Multiwall sacks.*—Multiwall sacks shall conform to the requirements of Federal Specification UU–S–48. Upon approval of the procuring agency, contractors may use their own commercial multiwall sacks if they are equal to or surpass sacks meeting the requirements of Federal Specification UU–S–48.

5.2.2.4 *Steel containers.*—Calcium chloride shall be packed in steel containers as specified in 5.2.1.3.

5.2.2.5 *One-pound and five-pound containers.* — One-pound and five-pound containers shall be packed in accordance with the domestic requirements of Federal Specification U–M–186.

5.2.3 *Level C.*—The calcium chloride shall be delivered in moistureproof bags containing not more than 100 pounds each, or in airtight drums weighing not more than 700 pounds each, or in bulk in tank cars, covered hopper cars, or covered trucks.

5.2.3.1 *One-pound and five-pound bottles.*—Twenty-four one-pound, or four five-pound bottles shall be packed in standard commercial shipping containers of the type, size, and kind commonly used for the purpose in a manner which will insure acceptance and safe delivery at destination. Shipping containers shall comply with the Uniform Freight Classification rules or other regulations as applicable to the mode of transportation.

5.2.4 *Standard pack (Civil agencies).*—One hundred pounds (net) of the product specified shall be packaged and packed in a multiwall paper sack equal to or better than multiwall paper sacks conforming to the requirements of Federal Specification UU–S–48, Grade L/W, sack construction number 14 L/W, as specified in table II of subject specification. Closure shall be with dipped ends.

5.3 Marking.

5.3.1 *Civil agencies.*—In addition to any special markings required by the contract or order, marking of shipments shall be in conformance with Federal Standard No. 123.

5.3.2 *Military agencies.*—In addition to any special markings required by the contract or order, marking of shipments shall be in conformance with Military Standard MIL–STD–129.

6. NOTES

6.1 Intended uses. — Type I and type II, grade A calcium chloride are intended for use for road purposes, as admixtures for concrete, as refrigerating mediums, and other technical uses. Type I calcium chloride is also intended for use as an antifreeze charge for water-type fire-extinguishers. Type II, grade B calcium chloride is intended for use as a desiccating (drying) agent. Calcium chloride should not be used as an antifreeze material in the water-cooling system of internal combustion engines.

6.2 CAUTION: Anhydrous calcium chloride is a very hygroscopic material and should be stored in tightly closed moisture-proof containers.

6.3 Ordering data.—Purchasers should exercise any desired options offered herein and should include the following information in procurement documents:

(*a*) Title, number, and date of this specification.

(*b*) Type, grade, and class of material required (see 1.2.1).

(*c*) Unit quantities and levels of packaging and packing required (see 5.1 and 5.2).

6.4 Supersession data.—This specification supersedes Interim Federal Specification O–C–00104a (GSA–FSS), dated July 16, 1956 and Federal Specification O–C–104, dated November 4, 1952, covering Calcium Chloride, Anhydrous, Technical Grade, and Interim Federal Specification O–C–00106b, dated Jan-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:12Z
Check the source to verify that this is the current version before use.

uary 11, 1957 and Federal Specification O–C–106a, dated March 1, 1944, covering Calcium Chloride, Dihydrate, Technical.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—C F M O Q
Navy—Y Or Sh S MC
Air Force.

Copies of this specification may be purchased for 5 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:12Z
Check the source to verify that this is the current version before use.